

Statistical versus Deep Learning Models for Post-Inauguration Bridge Traffic Forecasting: A Significance-Tested Comparison with Verification and Toll-Capacity Implications, Padma Bridge, Bangladesh

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ABSTRACT

Feasibility-stage traffic forecasts for megaproject bridges are rarely verified once the bridge opens, and no South Asian megaproject bridge has been checked against its original projections. This study provides that verification for the Padma Multipurpose Bridge, Bangladesh, using 1,442 days of Bangladesh Bridge Authority toll records (July 2022 to June 2026). Because operators need reliable day-ahead demand yet debate whether classical or deep learning methods serve better, a Seasonal Autoregressive Integrated Moving Average model with exogenous regressors (SARIMAX) and a two-layer Long Short-Term Memory (LSTM) network were fitted on 731 days and tested on 711 held-out days under one-step-ahead rolling forecasting, comparing accuracy through the Diebold-Mariano test, cross-validation, and five random seeds. The LSTM achieved lower holdout error (2,527 versus 2,814 vehicles per day root mean square error), yet the test could not reject equal accuracy ($p = 0.109$), while SARIMAX was unbiased and better cross-validated, making the two statistically indistinguishable. Both degraded within festival windows, amplifying error 3.90 and 2.70 times; pre-festival traffic averaged 35,991 vehicles per day against 27,729 after, an asymmetry whose peaks already exceed toll capacity. Observed traffic exceeded projections by 22 to 35 percent and motorcycles reached 17.5 percent of flow against a 1.9 percent forecast, divergences confirmed as genuine diverted demand through surveys of waterway and bus operators. These findings let authorities adopt either model on interpretability grounds and justify, from the observed peaks themselves, electronic toll conversion, a permanent motorcycle lane, and directionally scheduled festival capacity.

Keywords: Post-Inauguration Verification; Bridge Traffic Forecasting; SARIMAX; Long Short-Term Memory; Diebold-Mariano Test; Festival Demand Surge; Modal Shift; Toll Plaza Capacity; Padma Bridge; Bangladesh

1 Introduction

Megaproject bridges are justified before they are built. Traffic demand forecasts prepared years or decades in advance determine toll structures, financing terms, and the capacity of the toll plazas that will serve the facility for its design life. Once the bridge opens, those forecasts are rarely revisited, producing a persistent asymmetry: a substantial literature on how to forecast bridge traffic, and almost none on whether such forecasts held. Where retrospective evaluation exists, it is typically confined to development-bank completion reports rather than peer-reviewed analysis, and it seldom examines whether the vehicle mix, the temporal structure of demand, or the extremes that determine capacity behaved as projected.

The Padma Multipurpose Bridge offers an unusually clean case for closing this gap. At 6.15 kilometers it is the longest bridge in South Asia. It connects Dhaka to seventeen districts of Barishal and Khulna Divisions, home to roughly 21 million people, and it opened in June 2022, nine years later than the 2013 date assumed in its feasibility study. That delay is analytically valuable rather than merely unfortunate: the forecast was prepared for a Bangladesh with 0.64 cars per thousand people and delivered to a Bangladesh in which motorcycle ownership had grown for over a decade (ADB, 2010). Any divergence between projection and outturn is therefore informative about how quickly motorization reshapes the composition of induced demand.

The Bangladesh Bridge Authority publishes daily toll reports recording vehicle counts at the Mawa-end and Jajira-end plazas. Summed across both, these yield a continuous record of two-way daily traffic from the day of opening. This study assembles 1,442 consecutive daily observations spanning 1 July 2022 to 11 June 2026 and uses them to answer three questions the literature leaves open. First, does post-inauguration bridge traffic follow patterns regular enough to forecast, and does a classical statistical model or a deep learning model capture them better? This is not merely a horse race: statistical models impose interpretable, reproducible structure, whereas neural networks learn structure from data at the cost of interpretability and determinism, and which tradeoff a toll authority should accept depends on whether the accuracy difference is large enough to matter. That question has been asked repeatedly and answered inconsistently, largely because studies report point error without testing whether differences are statistically distinguishable from zero.

Second, where do both model classes fail, and is the failure structural or incidental? Bangladesh observes two annual Eid festivals whose dates migrate roughly eleven days earlier each solar year, producing demand surges that arrive at different calendar points every year. If errors concentrates in these windows, the operative planning problem is event-period accuracy, not average accuracy. Third, how far did observed demand diverge from the projections that justified the investment, and does the direction of divergence transfer to comparable projects?

The contribution is fourfold: (i) the first peer-reviewed post-inauguration traffic verification for a South Asian megaproject bridge, benchmarking 1,442 days of observed demand against feasibility projections at aggregate and vehicle-class level; (ii) the first application of the Diebold-Mariano test to a SARIMAX-versus-LSTM comparison on a multi-year daily bridge series, establishing that the two model classes are statistically indistinguishable despite differing point estimates; (iii) identification and quantification of pre- and post-festival demand asymmetry as the dominant residual error source, with a diagnosis of why binary event indicators cannot capture it; and (iv) an independent modal-shift validation drawing on primary field surveys and waterway operator data, confirming that observed growth reflects genuine diverted demand rather than modeling artifact.

2 Literature Review and Research Gap

2.1 Statistical and Deep Learning Traffic Forecasting

The autoregressive integrated moving average family, formalized by Box and Jenkins (1970), remains the reference framework for univariate traffic forecasting, characterizing a series through autoregressive terms on past values and moving-average terms on past errors, with differencing to induce stationarity. Ahmed and Cook (1979) established its competitiveness on freeway volumes, and the seasonal extension became standard for daily series with weekly cycles (Williams et al., 1998), remaining effective under sparse data (Kumar and Vanajakshi, 2015). Two properties

recommend it for deployment: every parameter is interpretable, and maximum-likelihood estimation via the Kalman filter is deterministic, returning identical parameters on every run. The exogenous extension conditions forecasts on known future events such as holidays.

Long short-term memory networks (Hochreiter and Schmidhuber, 1997) maintain a gated memory state that selectively retains information across time steps, overcoming the vanishing-gradient limitation of earlier recurrent networks. Zhao et al. (2017) showed LSTM outperforming classical methods on urban speed data, with comparable results reported for other traffic series (Farhan and Ong, 2018). Recent comprehensive reviews confirm a broad shift from statistical to data-driven traffic models while cautioning that deep networks demand more data and careful tuning (Afandizadeh et al., 2024). Critically, neural results depend on stochastic weight initialization, so identical code returns different accuracy on each run (Bouthillier et al., 2021); a single LSTM run against a single SARIMAX run compares a random draw against a fixed point. This study therefore reports LSTM performance across five seeds.

2.2 Comparative Studies and the Absent Significance Test

Direct SARIMA-versus-LSTM comparisons are now common and their results conflict. Ma et al. (2015) found the statistical model superior on short series but the neural model competitive on longer ones with structural breaks. Mohammadzadeh et al. (2023), forecasting daily rural traffic, found SARIMA weak at capturing irregularities and neural models prone to unrealistic spikes near national holidays, a pattern this study also observes. The interpretive difficulty is not incidental: forecast errors from two models on the same series are paired, and their loss differential is itself a time series with variance. Diebold and Mariano (1995) provide the appropriate test of equal expected loss, yet to our knowledge no published SARIMAX-versus-LSTM study on multi-year daily bridge traffic has applied it, so reported differences are routinely read as superiority when they may fall within sampling variation.

2.3 Megaproject Verification and Festival Demand

Flyvbjerg et al. (2006), examining 183 road projects, documented systematic optimism bias with actual traffic averaging 9.5 percent below forecast, though with very high variance. For bridges in motorizing economies the bias frequently reverses once induced demand and modal shift materializes. Yet verification studies for individual megaprojects remain rare, and where the feasibility document is public, as for the Padma Bridge (ADB, 2010; Raihan and Khondker, 2010), the comparison has not been performed. Festival demand adds a further complication: the Islamic lunar calendar advances about eleven days each solar year, so Eid migrates across months, and a fixed monthly seasonal factor will misattribute festival demand as the festival moves. Binary flags avoid this drift but impose a symmetric, constant effect on every window regardless of travel direction or bridge maturity, representing neither the outbound-inbound asymmetry nor the growth of festival peaks.

2.4 Waterway Displacement Following Bridge Construction

Across fixed-link (i.e. bridge) megaprojects worldwide, a consistent finding emerges, a new bridge rarely coexists with the ferry crossing it parallels as an added option, but instead displaces it, often within a short period of time. In Denmark, the Great Belt Bridge opened in 1998, replaced the ferry service that had previously linked Zealand to the rest of the country, and subsequent accessibility-based evaluations of the bridge's economic effects treat the ferry's withdrawal as the defining shock driving the observed productivity and wage responses in the regions it connected (De Borger, Mulalic, and Rouwendal, 2025). Similarly, in Canada, where the Confederation Bridge, opened in 1997, ended the century-old Cape Tormentine–Borden ferry crossing between Prince Edward Island and the mainland outright rather than merely competing with it (Canadian Encyclopedia, 2016). Around the Pearl River Delta, survey on the Hong Kong–Zhuhai–Macao Bridge also found that a large share of cross-boundary travelers reported using ferry terminals markedly less often after the bridge opened, attributing the shift to the bridge's speed and reliability advantage over the sea crossing it paralleled (Lam and Lei, 2023). Moreover, in Washington State, longitudinal analysis of the state ferry system, presented to a Transportation Research Board standing committee on ferry transportation, documents a sustained ridership decline as competing road and transit options improved relative access over time, underscoring that waterway demand is highly sensitive to the availability of faster land-based alternatives

even without a direct fixed-link replacement (Hough and Jin, 2019). The Padma Bridge in Bangladesh exhibits the same pattern locally. The bridge's effects on the Barisal Division transportation system found passenger ferries experiencing a sharp decline in passenger numbers and daily trips following the bridge's opening, with operators reducing services and staff losing employment as travelers shifted to the bridge and the bus routes it enabled (Rashid et al., 2023). Such transfer happens rapidly rather than gradually; it produces a structural break in the traffic series near the opening date that a purely seasonal statistical model or a neural network trained mostly on post-opening data will not represent without an explicit intervention term.

2.5 Research Gap

This study aims to fulfil three primary gaps identified in existing literature. First, although the Padma Bridge's feasibility document is publicly available, no peer-reviewed study has verified its post-inauguration traffic against the original pre-construction projections, and in particular, the extent to which the ferry-to-bridge modal displacement documented internationally, and reported informally for the Padma Bridge itself, corresponds to the diversion assumptions embedded in that forecast has not been established. Second, while comparisons of SARIMAX and LSTM models on bridge traffic are increasingly common, none has tested whether the resulting accuracy differences are statistically distinguishable from zero, nor quantified the run-to-run variability inherent to neural models across multiple random seeds. Third, the structure of festival-related forecasting error remains undiagnosed, leaving it unclear whether such error arises primarily from misestimation of festival demand magnitude or from unmodeled asymmetries between outbound and inbound travel directions.

3. Study Area, Data and Methodology

3.1 Study Area and Target Variable

The Padma Multipurpose Bridge carries a four-lane highway across the Padma River between Mawa in Munshiganj and Jajira in Shariatpur, providing the only fixed land link between Dhaka and the seventeen districts of Barishal and Khulna Divisions. Toll plazas operate at both ends. Before inauguration, road traffic crossed by vehicle ferry at Shimulia-Banglabazar or Paturia-Daulatdia, and passengers travelled predominantly by overnight launch from Sadarghat terminal. The target variable is total two-way daily traffic, the sum of Mawa-end and Jajira-end counts on the same calendar day, which the Authority reports as the daily headline figure and which is therefore the natural basis for comparison against official projections.

3.2 Data Collection

This study draws on three data streams, each assembled through a distinct process and reconciled before analysis. The primary traffic data came directly from the Bangladesh Bridge Authority through a formal institutional request initiated by BUET and processed under the Authority's standard data-release procedure. The records were provided in two stages, in 2023 and 2024, as daily class-wise vehicle counts, from which daily total two-way traffic was compiled. The 2022 to 2024 records supported model specification and accuracy checking, while the 2025 to 2026 records were reserved as an independent hold-out set, used to validate the forecasts and to confirm consistency against the earlier data.

The modal-shift evidence was gathered through primary field surveys over eight clear-weather days at the Sadarghat launch terminal, the Bangladesh Inland Water Transport Authority (BIWTA) office, and the Sayedabad bus terminal, covering roughly 150 respondents. On the waterway side there included around 30 launch passengers and staff, some 15 to 17 launch owners and managers, and several BIWTA officials. On the roadside, the survey spanned 22 bus companies operating southern routes, interviewing owners, drivers, ticket managers, and helpers alongside passengers in the waiting areas, capturing both operator and passenger perspectives on fares, service frequency, occupancy, and mode choice before and after the bridge opened. These field data were then supplemented with current online reports and published literature to extend the record beyond the survey period and corroborate the findings. All three streams were finally cleaned, checked for internal consistency, and reconciled into the analysis dataset.

The full record covers 1,442 consecutive days from 1 July 2022 to 11 June 2026 (BBA, 2026). One observation, 31 March 2026, was absent from the source and imputed as the mean of the adjacent days; all other 1,441 values are direct readings. The series has a mean of 18,293 vehicles per day and a standard deviation of 5,174, ranging from 2,673 to 52,487. Vehicle-class shares from a 183-day ground-truth subsample (June 2023 to January 2024) are motorcycle 17.5 percent, private vehicles 35.0 percent, bus 22.0 percent, and truck classes 18.8 percent. Two annual Eid festivals fall within the record on nine occasions; because the lunar calendar advances about eleven days yearly, these migrate across the Gregorian calendar. Each event is represented as a seven-day window (three days before, the day itself, three days after), yielding 63 flagged days (Table 1).

Table 1. Variables, Definitions, and Data Sources

Variable / data	Type	Definition	Source
Total traffic y_t	Continuous	Sum of Mawa-end and Jajira-end vehicle counts, day t	BBA (2026) toll report, formal two-stage release (2023, 2024)
Class-wise counts	Continuous	Daily counts by 13 vehicle classes	BBA (2026); 183-day ground-truth window
Eid window	Binary	1 on 3 days before, day of, 3 days after each of 9 events	Islamic calendar; authors' construction
Modal-shift indicators	Mixed	Launch fares, occupancy, trips; bus frequencies	Authors' primary field survey (2022-2024), ~150 respondents
Corroborating records	Text/numeric	Passenger volumes, operator income, booth counts	News and operator reports (2022-2026); see References
Feasibility projection	Continuous	Forecast AADT and class volumes	ADB (2010), Table 2-1; Raihan and Khondker (2010)

The record is partitioned chronologically: days 1 to 731 (July 2022 to June 2024) for training and specification, and days 732 to 1,442 (July 2024 to June 2026) as 711 out-of-sample validation days that no model saw during estimation. This partition is strict.

3.3 Stationarity Assessment

Autoregressive and moving-average terms require a stationary series, so differencing orders are established first. The Augmented Dickey-Fuller test takes a unit root as its null (Dickey and Fuller, 1979); the Kwiatkowski-Phillips-Schmidt-Shin test takes stationarity as its null (Kwiatkowski et al., 1992). Agreement across both is stronger evidence than either alone. On the raw series the two disagree, characteristic of a series mean-reverting around a slowly moving level, exactly what a bridge in demand ramp-up produces. First differencing resolves the conflict, seasonal differencing at lag seven likewise, and both together give the strongest statistic; the specification ($d = 1$, $D = 1$, $s = 7$) is adopted (Table 2).

Table 2. Stationarity Test Results (full 1,442-day series)

Transformation	ADF stat	ADF p	KPSS stat	KPSS p	Verdict
Raw y_t	-5.754	<0.001	1.949	<0.010	Conflicting: mean-reverting around moving level
First difference (d=1)	-11.484	<0.001	0.085	>0.100	Both agree: stationary
Seasonal difference (D=1, s=7)	-10.205	<0.001	0.010	>0.100	Both agree: stationary
Both (d=1, D=1, s=7)	-13.761	<0.001	0.006	>0.100	Adopted specification

The interpretation matters for the modeling choice that follows. A series that is stationary only after both a first difference and a seasonal difference carries two distinct sources of persistence: a slow level drift reflecting demand growth, and a weekly cycle reflecting the Friday holiday. Removing both leaves innovations that the autoregressive and moving-average terms can model.

3.4 SARIMAX Specification

The seasonal ARIMA model with exogenous regressors treats the series as a linear regression on external predictors whose residual follows a seasonal ARIMA process. With B the backshift operator ($By_t = y_{t-1}$):

$$y_t = \beta^T x_t + u_t, \quad \varphi_p(B)\Phi_p(B^s)(1-B)^d(1-B^s)^D u_t = \theta_q(B)\Theta_q(B^s)\varepsilon_t \quad (1)$$

Here y_t is total two-way traffic on day t and x_t is the exogenous regressor vector containing the Eid-window indicator and the Year 2 indicator, with β the associated coefficient vector. The residual u_t follows the seasonal ARIMA process on the right. The non-seasonal autoregressive and moving-average polynomials are $\varphi_p(B) = 1 - \varphi_1 B - \dots - \varphi_p B^p$ and $\theta_q(B) = 1 + \theta_1 B + \dots + \theta_q B^q$; the seasonal polynomials Φ_p and Θ_q take the same form in B^s with $s = 7$. The orders d and D are the non-seasonal and seasonal differencing orders established in Section 4.1, and ε_t is a white-noise innovation with variance σ^2 .

Two features of this specification require justification. The seasonal period is set to seven days rather than 365 because the weekly cycle is the only fixed-period seasonality present. Annual seasonality in this series is driven by Eid, whose date migrates, so a 365-day seasonal term would misattribute festival demand to fixed calendar positions and would in any case be unidentifiable from four years of data. Eid is instead handled through the exogenous indicator, which travels with the actual festival date.

The year indicator merits equal care. Four operational years exist in the full record, but only Years 1 and 2 fall within the 731-day training block. Indicators for Years 3 and 4 are therefore structurally zero in-sample and perfectly collinear with the intercept; including them yields undefined standard errors. Only the Year 2 indicator is estimable and only it is included. This is a limitation of the training-block length rather than a modeling choice, and it means the model must extrapolate the level shift into Years 3 and 4 rather than estimate it. We return to this point when interpreting forecast bias.

Model orders are selected by comparing information criteria across nested specifications, reported in Section 4. Parameters are estimated jointly by maximum likelihood via the Kalman filter, which propagates the state vector forward through the training block and identifies the parameter vector maximizing the likelihood of the observed sequence.

3.5 Long Short-Term Memory Specification

The LSTM reads a sliding window of $L = 28$ days (four weekly cycles) and predicts the next day. At each step the cell updates memory through three sigmoid-activated gates:

$$f_t = \sigma(W_f [h_{t-1}, x_t] + b_f), \quad i_t = \sigma(W_i [h_{t-1}, x_t] + b_i), \quad o_t = \sigma(W_o [h_{t-1}, x_t] + b_o) \quad (2)$$

The forget gate f_t sets how much previous cell state to retain, the input gate i_t how much candidate information to write, and the output gate o_t how much to expose. The cell and hidden states update as:

$$\tilde{c}_t = \tanh(W_c [h_{t-1}, x_t] + b_c), \quad c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad h_t = o_t \odot \tanh(c_t) \quad (3)$$

where σ is the sigmoid, $\odot$ elementwise multiplication, and W, b are learned. The final hidden state passes through a dense layer giving $\hat{y} = W_y h_L + b_y$. The input vector at each step contains eleven features: the standardized traffic value, the Eid-window indicator, seven day-of-week indicators, and the Year 2 indicator. Standardization uses the training-block mean and standard deviation only, preventing any leakage of validation-period information into the scaling. The architecture stacks a 64-unit LSTM layer beneath a 32-unit layer, with dropout at rates 0.2 and 0.1 respectively; the hierarchy allows the first layer to extract short-range temporal features and the second to integrate them. Training uses the Adam optimizer with learning rate 0.0008, batch size 32, and up to 150 epochs with early stopping after 15 epochs without improvement and learning-rate halving after 7.

Because network weights are randomly initialized, a single training run represents one draw from a distribution of possible models. The network is therefore trained five times under independent random seeds and the full distribution of validation performance is reported. This is not a robustness afterthought: without it, any comparison against the deterministic SARIMAX estimate is uninterpretable.

3.6 Evaluation Protocol and Comparative Testing

Both models are evaluated over the 711-day validation block under a one-step-ahead rolling protocol. On each validation day the model forecasts the following day, the true observation is then revealed, and the model state is updated before the next forecast. For SARIMAX the Kalman filter state is updated analytically without re-estimating parameters. For the LSTM the update is implicit: the 28-day window advances by one day, replacing its oldest observation with the newly revealed value. This protocol mirrors operational deployment, in which yesterday's count is known before today's forecast is required, and it ensures neither model benefits from information unavailable at forecast time.

Accuracy is reported through root mean square error, mean absolute error, mean absolute percentage error, and the coefficient of determination. Two baselines contextualize these figures: a random walk forecasting y_{t-1} and a seasonal naive forecasting y_{t-7} . A model that cannot beat the seasonal naive has not learned anything the calendar does not already supply.

The central comparative question is whether the difference between SARIMAX and LSTM accuracy is distinguishable from sampling variation. Let e_{1t} and e_{2t} denote the squared forecast errors of the two models on day t and let $d_t = e_{1t} - e_{2t}$. Under the null hypothesis of equal expected loss, the mean of d_t is zero. The Diebold and Mariano (1995) statistic tests this directly:

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})}}, \quad \text{where } \bar{d} = \frac{1}{n} \sum_t d_t \quad (4)$$

and $\hat{V}(\bar{d})$ is a consistent estimator of the variance of the loss differential. Under the null the statistic is asymptotically standard normal. For one-step-ahead forecasts the loss differential is serially uncorrelated at the relevant lag, so the sample variance is consistent.

Generalization beyond the single validation block is assessed by five-fold expanding-window cross-validation. Standard k-fold cross-validation randomly permutes observations and would permit a model to train on future data to predict the past, which is invalid for time series. The expanding-window variant requires each fold's training set to precede its test set chronologically, and both models are re-estimated from scratch on every fold.

4 Data Diagnostics and Model Adequacy

A model can achieve low in-sample error while violating the assumptions that make its inference valid, so each diagnostic is tested and reported, including results that do not favor the adopted specification.

4.1 Model Order Selection

Three nested specifications are compared on the training block (Table 3): Model A with no exogenous regressors, Model B adding the Eid indicator, Model C adding the Year 2 indicator. Adding the Eid indicator improves log-likelihood by 10.8 at one parameter, cutting the Akaike criterion by 19.6 and the Bayesian by 15.0, a decisive gain confirming festival demand is separately identifiable. Model C does not improve on B by either criterion, an honest and instructive result: the Year 2 coefficient is large and significant when estimated (7,811 vehicles per day, standard error 1,393, $p < 0.001$), yet seasonal differencing already removes much of the level shift, so the indicator explains variance the model has partly eliminated. We nonetheless retain Model C for forecasting because the validation block lies beyond Year 2 and the level-shift information is required for extrapolation, not in-sample fit, a defensible but not uncontested choice whose effect on validation error proved negligible.

Table 3. Model Comparison on the 731-Day Training Block

Model	Exog.	Log-lik.	AIC	BIC	Preferred by
A: SARIMA (2,1,2) (1,1,1)₇	None	-6,617.2	13,248.4	13,280.4	—
B: + Eid window	1	-6,606.4	13,228.8	13,265.4	AIC and BIC
C: + Eid + Year 2	2	-6,607.2	13,232.3	13,273.4	(retained for forecasting)

4.2 Residual Diagnostics

Three findings warrant discussion rather than assertion. The Ljung-Box test passes at lags 7, 14, and 28 but marginally fails at lag 21 ($p = 0.045$). We report this rather than selecting a lag at which the model passes. A failure at a single intermediate lag, bracketed by passes on either side, is more consistent with sampling variation across four correlated tests than with a genuine three-week dependency; under a Bonferroni correction for four tests the critical value would be 0.0125 and the lag-21 result would pass comfortably. Nevertheless, the possibility of residual three-week structure cannot be excluded, and identifying its source would be a worthwhile extension, as summarized in Table 4.

The Breusch-Pagan test decisively rejects homoscedasticity, which Figure 1(a) and 1(c) make visually obvious: residual variance grows with the fitted value. This is expected in a traffic series whose absolute variability scales with volume, and it is not a defect of the model but a property of the data. Its consequence is that forecast intervals are too narrow at high volumes, precisely where festival days lie. Point forecasts remain unbiased.

The Jarque-Bera test rejects normality with excess kurtosis of 13.1. Figure 1(b) locates the source: the tails depart from the normal reference at both extremes, and the departing observations are festival days. A model whose residuals are normal on ordinary days and heavy-tailed on festival days is telling the analyst that festival days are not drawn from the same distribution as ordinary days. That is a substantive finding about the demand process, and Section 6.4 pursues it.

Taken together the diagnostics support the adopted specification for point forecasting while identifying two qualifications: interval estimates at high volumes are optimistic, and festival days are imperfectly represented. Both are stated in conclusions rather than buried.

Table 4. Residual Diagnostic Summary (SARIMAX Model C, $n = 721$)

Test	Statistic	Null hypothesis	Result	Consequence
Ljung-Box (7)	$Q=2.27, p=0.943$	Residuals white noise	PASS	Weekly structure captured No action
Ljung-Box (14)	$Q=21.96, p=0.079$	Residuals white noise	PASS	

Ljung-Box (21)	Q=33.15, p=0.045	Residuals white noise	MARGINAL	Reported openly
Ljung-Box (28)	Q=36.49, p=0.131	Residuals white noise	PASS	Not systematic
Durbin-Watson	1.933	No 1st-order autocorr.	PASS	Acceptable
Breusch-Pagan	LM=79.4, p<0.001	Homoscedastic	REJECT	Intervals widen at volume
Jarque-Bera	JB=3130, p<0.001	Normal	REJECT	Eid tails (skew 0.59, kurt 13.1)
Residual mean	-58.1	≈ 0	PASS	0.3% of mean; unbiased

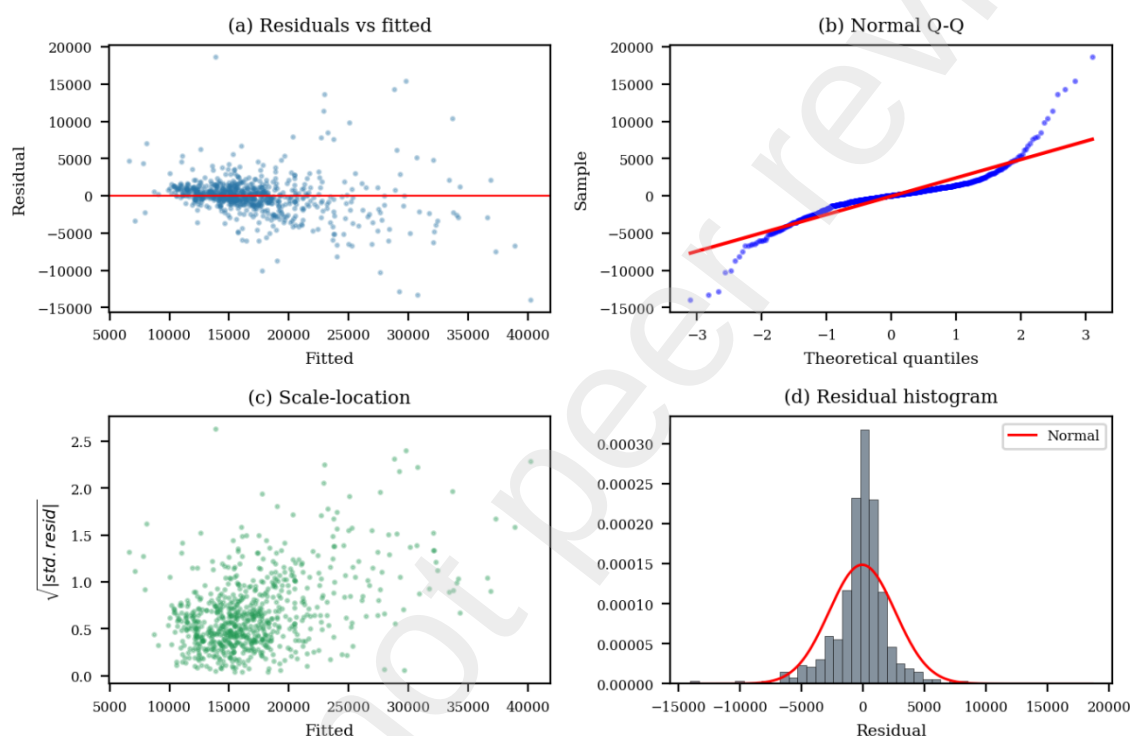


Figure 1. SARIMAX residual diagnostics: (a) residuals versus fitted showing variance growth; (b) normal Q-Q with heavy tails; (c) scale-location confirming heteroscedasticity; (d) histogram against a normal density. Departures concentrate in festival-period observations.

5 Results and Discussion

5.1 Observed Demand Structure

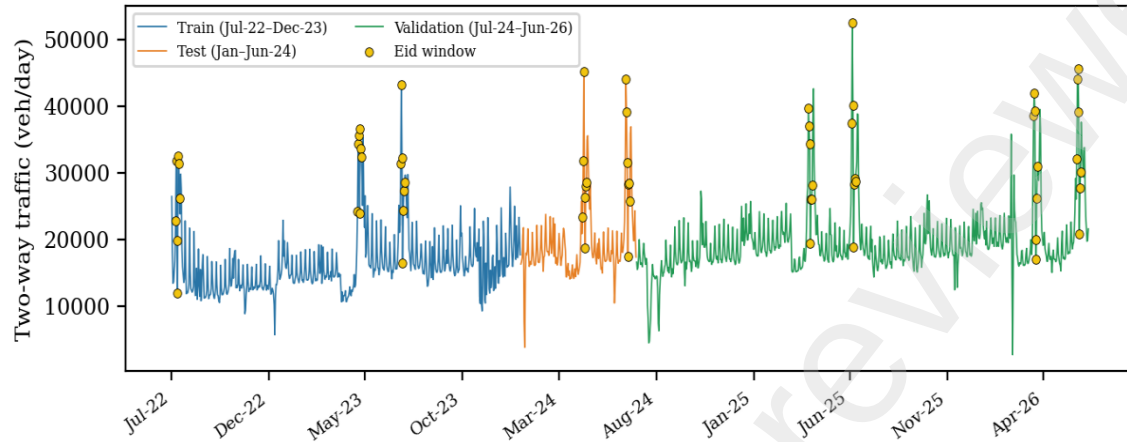


Figure 2. Daily two-way traffic, July 2022 to June 2026 ($n = 1,442$). Colors mark the training, test, and validation partitions; gold markers denote the 63 Eid-window days.

Annual average daily traffic rose from 15,686 vehicles per day in Year 1 to 19,091 in Year 3, a compound annual growth rate of 10.3 percent, slowing to 20,045 in the partial fourth year. The deceleration from 17.6 percent year-on-year growth between Years 1 and 2 to 3.5 percent between Years 2 and 3 signals a facility completing its ramp-up (Table 5).

Table 5. Annual Average Daily Traffic by Operational Year

Year	Period	Days	AADT	YoY growth
Year 1	Jul-2022 to Jun-2023	365	15,686	— (base)
Year 2	Jul-2023 to Jun-2024	366	18,440	+17.6%
Year 3	Jul-2024 to Jun-2025	365	19,091	+3.5%
Year 4 (partial)	Jul-2025 to Jun-2026	346	20,045	+5.0%

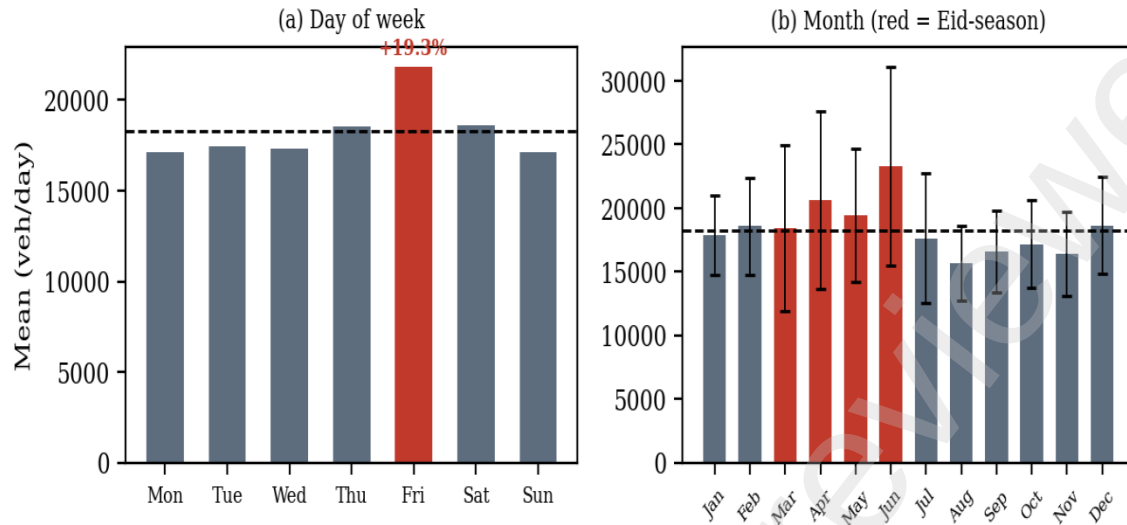


Figure 3. (a) Mean traffic by day of week; Friday carries 21,824 vehicles per day, 19.3 percent above the series mean. (b) Mean traffic by month with error bars; red bars mark Eid-containing months. The March-June elevation is a festival-timing artifact, not fixed seasonality.

The weekly pattern is stable across all four years and motivates the period-seven seasonal term. The monthly pattern (in Figure 3) requires more careful interpretation. March through June appear elevated, and a naive reading would treat this as fixed seasonality. It is not. Eid-ul-Adha fell in July in 2022, June in 2023 through 2025, and May in 2026; Eid-ul-Fitr moved from April 2023 to March 2026. The monthly elevation therefore tracks the festivals rather than any property of the months themselves, and a monthly seasonal factor estimated on this record will mislead as the festivals continue to migrate. This is the reason the model handles Eid through a date-linked indicator rather than a calendar-month term, and it is a design consideration that transfers to any Muslim-majority country.

5.2 Forecast Accuracy and the Indistinguishability of Model Classes

Over the 711 validation days both models comfortably beat both naive baselines (Table 6), against the seasonal naive, SARIMAX cuts RMSE by 45.9 percent. This validation is itself the primary check on the models: forecasts are matched day-by-day against real observed traffic that no model saw during training, and the close tracking in Figure 4 is that comparison made visible. Read naively, the LSTM wins: its root mean square error is 287 vehicles per day lower and it explains 74.5 percent of validation variance against 68.4 percent. Two considerations complicate that reading, and both point in the same direction.

First, the SARIMAX forecast is effectively unbiased, with a mean error of -3 vehicles per day on a series averaging 18,293. The LSTM under-forecasts by 447 vehicles per day on average across seeds. The source is identifiable: the network's training block contains only Years 1 and 2, and its 28-day input window cannot signal that the series has entered a permanently higher regime. SARIMAX avoids this because differencing removes the level entirely, so it models change rather than levels and a permanent level shift does not displace it. A model that is more accurate on average but systematically low is a different operational proposition from one that is slightly less accurate but centered, particularly for revenue forecasting where bias accumulates.

Second, and decisively, the accuracy difference does not survive a formal test. The Diebold-Mariano statistic on the squared-error loss differential is 1.602, corresponding to $p = 0.109$. The null hypothesis of equal predictive accuracy cannot be rejected at any conventional level. The 287-vehicle gap in root mean square error, which would be reported as an LSTM victory in the format conventional in this literature, lies within sampling variation of the 711-day evaluation window.

Table 6. Validation Performance, 711 Out-of-Sample Days

Model	RMSE	MAE	MAPE	R ²	Bias
Random walk (y_{t-1})	3,297	—	10.81%	—	—
Seasonal naive (y_{t-7})	5,197	—	14.74%	—	—
SARIMAX + Eid + Year 2	2,814	1,537	8.11%	0.684	-3
LSTM, seed 42	2,527	1,491	8.53%	0.745	-486
LSTM, mean of 5 seeds	2,559±26	1,508±13	8.66±0.09%	0.739±0.005	-447±79

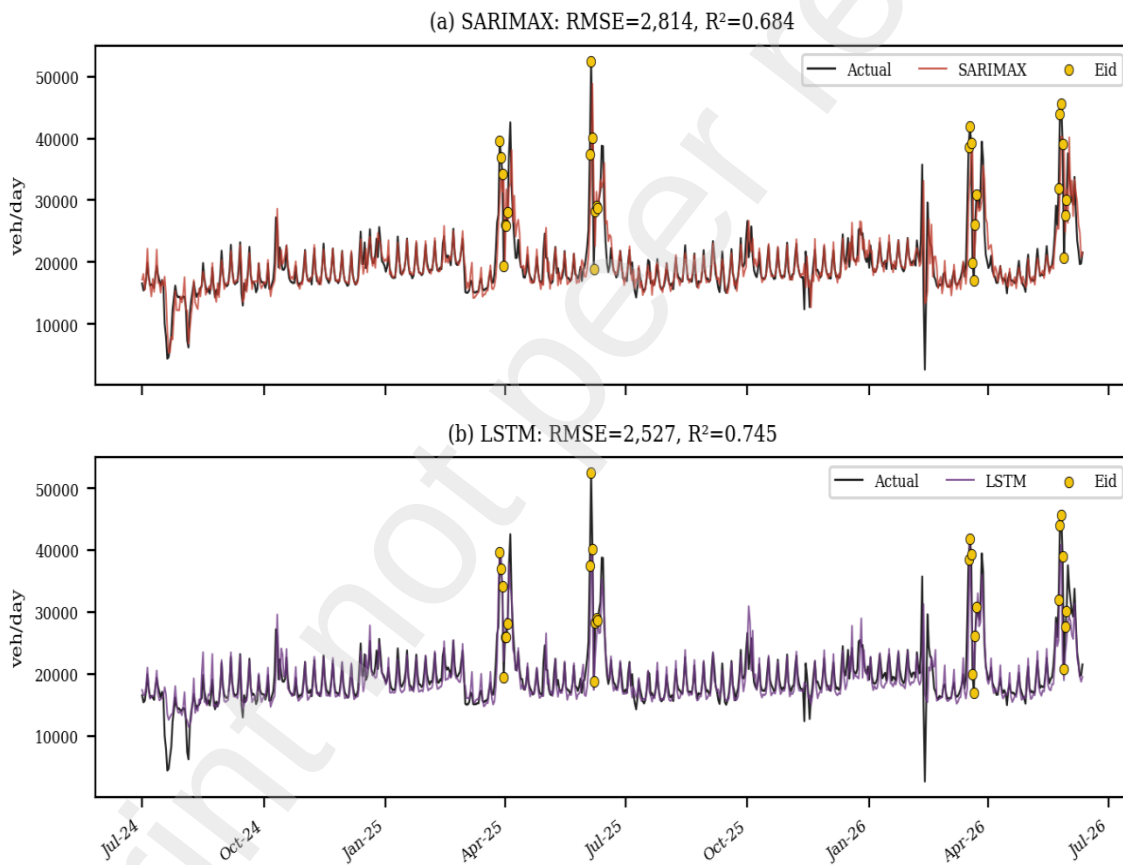


Figure 4. One-step-ahead forecasts against observed traffic over the 711 validation days: (a) SARIMAX; (b) LSTM. Both track ordinary variation closely and fall short at festival peaks.

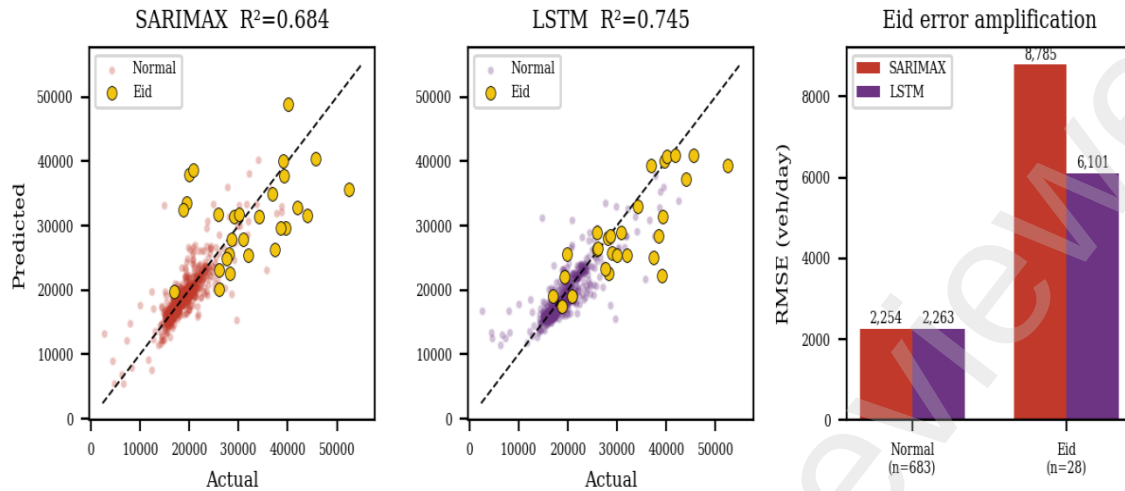


Figure 5. Predicted against observed for SARIMAX (left) and LSTM (center), identity line dashed; festival days cluster below the line for both. (right) RMSE by day type: both degrade sharply within Eid windows.

Cross-validation reverses the ordering: across five expanding-window folds SARIMAX attains mean RMSE $2,908 \pm 753$ against LSTM's $3,119 \pm 924$, the neural model performing poorly on early folds with short training windows and overtaking on Folds 4 and 5, consistent with a high-parameter model that improves as data accumulates (Figure 6).

The practical conclusion is not that one model is superior. It is that on this series, at this sample size, the two model classes are statistically indistinguishable in accuracy, and selection should therefore rest on other criteria (Figure 5 and 6). SARIMAX offers deterministic estimation, interpretable coefficients, and unbiased forecasts. The LSTM offers marginally lower point error on the full training block at the cost of run-to-run variability, though we note that variability proved modest here: across five seeds the root mean square error standard deviation was only 26 vehicles per day, or 1.0 percent of its mean, suggesting the reproducibility concern raised in the machine learning literature is less acute for this problem than for many others. For a toll authority whose forecasts must be defensible to auditors and regulators, interpretability and determinism plausibly dominate a 287-vehicle accuracy difference that no test can distinguish from zero.

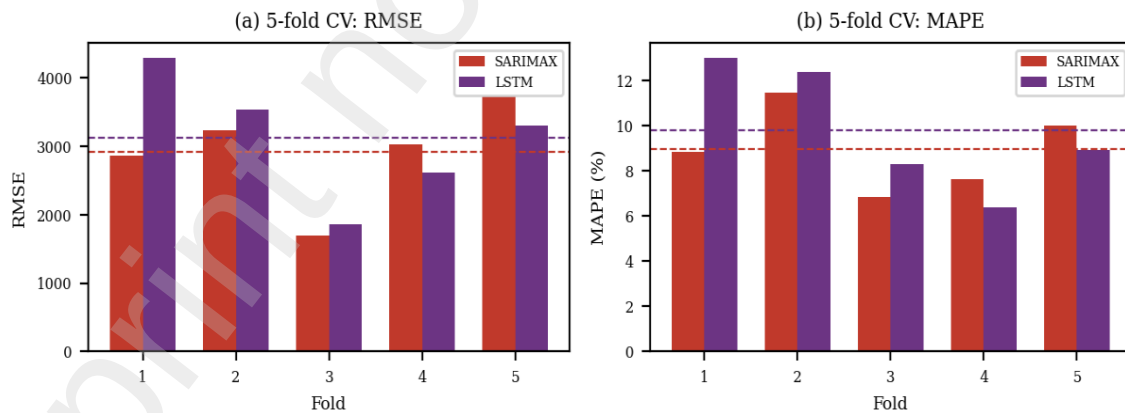


Figure 6. Five-fold expanding-window cross-validation; dashed lines mark fold means. LSTM underperforms on early folds and overtakes later ones as training data accumulate.

5.3 Where Both Models Struggle: Festival Windows

Disaggregating error by day type locates the failure precisely. On 683 ordinary validation days SARIMAX and LSTM attain RMSE of 2,254 and 2,263, essentially identical; on 28 Eid-window days these rises to 8,785 and 6,101, amplification of 3.90 and 2.70 times, with the LSTM's flexible form absorbing some surge that SARIMAX's linear

coefficient cannot. Both models under-forecast peaks (SARIMAX mean Eid error -880 , LSTM $-3,096$). Figure 7 shows the record peak, where both anticipate a surge and understate it.

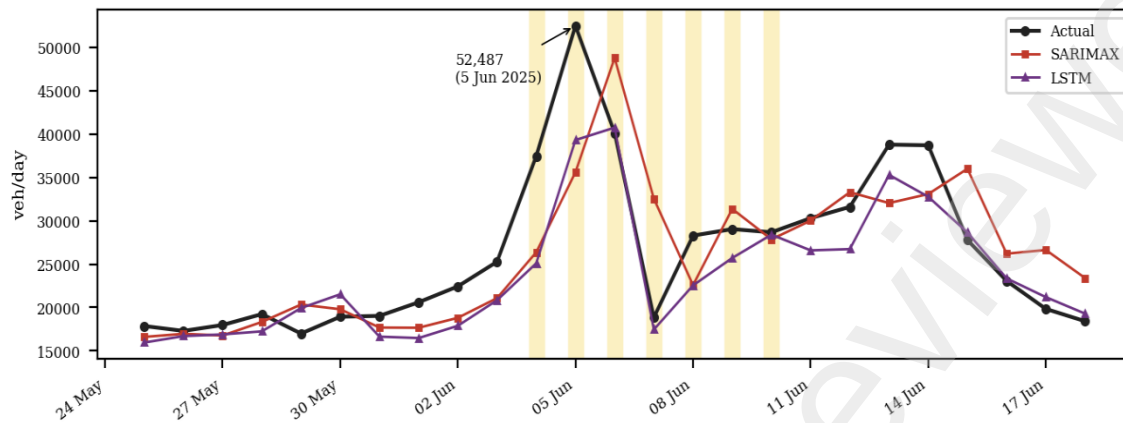


Figure 7. Eid-ul-Adha 2025. Observed traffic reached 52,487 vehicles on 5 June 2025, the record. Both models anticipate the surge and understate its magnitude.

5.4 Diagnosing the Festival Error: Asymmetry, Not Magnitude

The conventional explanation is that a binary indicator estimates one average effect and understates large events; if so, error would shrink as more festivals are observed. It does not, so the nine events are decomposed directly (Table 7). Two findings emerge. The first is negative: regressing the peak-to-mean ratio on bridge year gives a positive slope of 0.128 per year but with standard error 0.084 and $p = 0.173$ across nine events, so the apparent escalation from 1.77 to 2.87 times the mean is consistent with an upward trend but not established by it. We state plainly that year-over-year escalation is suggested by the data and not demonstrated, a natural subject for continued monitoring. The Fitr-versus-Adha difference is likewise not significant ($t = -0.683$, $p = 0.517$).

Table 7. Eid Event Decomposition (seven-day windows)

Event	Br. yr	Window mean	Peak day	Peak/mean	Pre-3	Post-3
Eid-ul-Adha 2022	1	25,165	32,440	1.77×	24,741	29,993
Eid-ul-Fitr 2023	1	31,455	36,519	2.00×	31,309	34,140
Eid-ul-Adha 2023	1	29,024	43,137	2.36×	35,526	26,721
Eid-ul-Fitr 2024	2	28,808	45,204	2.47×	33,437	25,045
Eid-ul-Adha 2024	2	30,621	44,033	2.41×	38,187	27,449
Eid-ul-Fitr 2025	3	30,044	39,637	2.17×	36,934	26,696
Eid-ul-Adha 2025	3	33,560	52,487	2.87×	43,357	28,666
Eid-ul-Fitr 2026	4	30,515	41,885	2.29×	39,894	24,668
Eid-ul-Adha 2026	4	34,174	45,602	2.49×	40,532	26,181

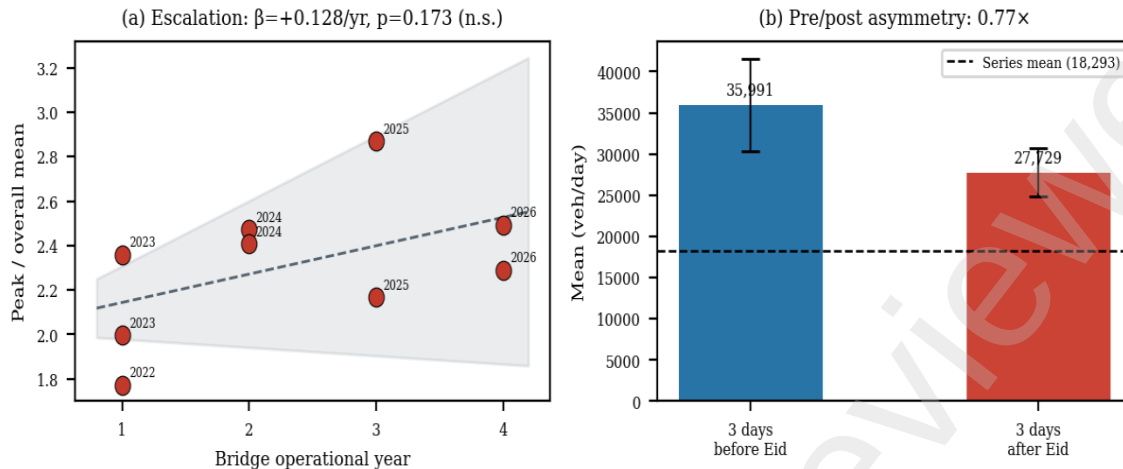


Figure 8. (a) Peak-to-mean ratio against bridge year; the fitted slope is positive but not significant ($\beta = +0.128/\text{yr}$, $p = 0.173$, $n = 9$). (b) Mean traffic three days before versus after each festival; the pre-festival level exceeds the post by a factor of 1.30.

The second finding is positive, robust, and to our knowledge new. Traffic in the three days before each festival averages 35,991 vehicles per day against 27,729 in the three days after, a 30 percent excess holding in seven of nine events. The mechanism is behavioral: outbound travel compresses into a narrow window before a fixed date, while return spreads over the following week. A binary indicator assigns the same coefficient to a day three days before Eid and a day three days after (Figure 8), so it is not merely too small but the wrong shape, and no accumulation of festivals will fix a symmetric regressor applied to an asymmetric process.

This diagnosis reframes the modeling problem. A binary indicator assigns the same coefficient to a day three days before Eid, when traffic averages 35,991, and a day three days after, when it averages 27,729. The indicator is not merely too small; it is the wrong shape. No amount of additional festival observations will fix a symmetric regressor applied to an asymmetric process. The appropriate response is a signed, directional event regressor, for instance a countdown variable taking values $-3, -2, -1, 0$ before and on the festival and a separate return-flow variable taking values $1, 2, 3$ after, allowing the model to estimate distinct build-up and dissipation coefficients. Because Eid dates are known from the Islamic calendar more than a year in advance, such regressors are fully available at forecast time and impose no operational cost. We did not implement this here because doing so would require re-specifying and re-validating both models, expanding the scope beyond the comparative question this paper addresses; it is the single most promising extension of this work and we identify it as such.

5.5 Design-Day Projection and Its Horizon Limits

A natural question is what traffic to expect over the next two years. Here the distinction between rolling and multi-step forecasting is decisive. The 8 percent accuracy reported above is a rolling one-step result, in which the true previous value is always available; a genuine multi-year forecast has no such feedback, and when the model is fed its own forecasts the error compounds, with the coefficient of determination turning negative within weeks and the 80 percent interval widening past plausibility. This is not specific to our data but inherent to daily time-series models, which are short-horizon instruments. We therefore do not extrapolate the daily model across two years. Instead, following standard practice in this literature, we project a seasonal design-day profile: an annual level grown at assumed rates, distributed by the four-year seasonal structure, with festivals placed on their migrating future dates.

Three growth scenarios bound the projection, at 4, 5, and 7 percent per year, with 5 percent the central case because it matches the observed settled Year 3-to-4 rate; the early 17.6 percent ramp-up is excluded as a one-time diversion effect. These rates are projections conditioned on assumptions, not model outputs, and the realized rate will depend on regional and national population growth, land-use change and urbanization along the corridor, industrial and agricultural development, fuel prices, and macroeconomic conditions. Rather than the monthly average, we report the design day, the single busiest day expected each month, since capacity is governed by peaks. Festival design days

apply a multiplicative factor of 2.35 to the baseline, the mean peak-day-to-baseline ratio observed across the nine events (Table 7); a multiplicative factor is used rather than the model's additive coefficient because the proportional festival lift was stable across years while the absolute lift grew with the baseline.

Table 8. Projected Design-Day Traffic by Month (max traffic on any single day; central 5%/yr, range 4-7%)

Month	FY2026-27 (5%)	Range 4-7%	Month	FY2027-28 (5%)	Range 4-7%
Jul-26	29,700	29,400-30,300	Jul-27	31,200	30,600-32,400
Aug	25,300	25,000-25,700	Aug	26,500	26,000-27,600
Sep	26,000	25,800-26,500	Sep	27,300	26,800-28,400
Oct	27,700	27,400-28,200	Oct	29,000	28,500-30,200
Nov	24,900	24,600-25,300	Nov	26,100	25,600-27,100
Dec	29,400	29,100-29,900	Dec	30,800	30,200-32,000
Jan-27	25,400	25,200-25,900	Jan-28	26,700	26,200-27,700
Feb	29,600	29,300-30,100	Feb (Eid)	51,900	50,900-53,900
Mar (Eid)	49,500	49,000-50,400	Mar	31,900	31,300-33,100
Apr	39,200	38,800-39,900	Apr	41,100	40,400-42,700
May (Eid)	49,500	49,000-50,400	May (Eid)	51,900	50,900-53,900
Jun	38,500	38,200-39,300	Jun	40,500	39,700-42,000

The Eid design days, near 49,500 in 2026-27 and 51,900 in 2027-28 (Table 8), approach and under the high scenario exceed the current record of 52,487, confirming that peak-day pressure will persist. Eid correctly migrates from March and May in 2027 to February and May in 2028, and April, elevated in earlier years only because Eid passed through it, returns to an ordinary level.

Class-wise projections apply class-specific growth from the observed settled rates, motorcycle 2 percent, private vehicles 6 percent, bus 4 percent, and truck 5 percent, so the blended total remains near 5 percent (Table 9). The motorcycle rate is deliberately modest: although motorcycles were underestimated roughly twelvefold by the feasibility forecast and reached a 17.5 percent share during the post-ban-lift surge captured in the ground-truth window, that surge has since plateaued, and the settled share is closer to 10 percent. Projecting the surge rate forward would repeat exactly the error the feasibility study made.

Table 9. Class-Wise Design-Day Projection (two-way veh/day; class-specific growth)

Class (fwd rate)	FY26-27 AADT	FY26-27 normal day	FY26-27 Eid day	FY27-28 AADT	FY27-28 normal day	FY27-28 Eid day
Motorcycle (2%)	2,170	2,755	5,098	2,213	2,810	5,200
Private (6%)	9,224	11,714	21,676	9,777	12,417	22,977
Bus (4%)	5,381	6,834	12,646	5,596	7,107	13,151
Truck/other (5%)	4,245	5,391	9,975	4,457	5,660	10,474
Total	21,019	26,694	49,395	22,043	27,995	51,802

6 Validation Against Feasibility Forecasts and Modal Shift

6.1 Observed Traffic Against Feasibility Projections

The feasibility study projected an opening-year AADT of 12,056 and a third-year figure of 15,587, assuming a 2013 opening (ADB, 2010; Raihan and Khondker, 2010). Mapping feasibility years to actual operational years, observed Year 1 traffic of 15,686 exceeded the projection by 30.1 percent and Year 3 of 19,091 by 22.5 percent, the induced-demand reversal documented for bridges in motorizing economies (Flyvbjerg et al., 2006). The class comparison is

starter (Table 10): the motorcycle projection underestimated observed volume roughly twelvefold, while bus and truck projections overstated volumes by 30 to 40 percent. The nine-year gap between a 2010 forecast and a 2022 opening spans the period of fastest motorcycle growth in Bangladesh, so the transferable lesson is that class-composition forecasts for long-delayed megaprojects in motorizing economies should treat the motorcycle share as the dominant uncertainty.

Table 10. Feasibility Forecast Against Observed Traffic by Vehicle Class

Vehicle Class	Year	Forecast	Observed est.	Ratio
Motorcycle	1	228	2,745	12.0×
Motorcycle	3	308	3,341	10.8×
Private	1	2,658	5,490	2.1×
Private	3	3,607	6,682	1.9×
Bus	1	5,693	3,451	0.61×
Bus	3	6,518	4,200	0.64×
Truck	1	3,477	2,949	0.85×
Truck	3	5,154	3,589	0.70×
Total AADT	1	12,056	15,686	1.30×
Total AADT	3	15,587	19,091	1.22×

Observed class volumes estimated by applying the 183-day ground-truth shares to each year's aggregate (ADB, 2010, Table 2-1).

6.2 Modal-Shift Validation of Observed Growth

If the observed growth reflects genuine diverted demand, the modes the bridge displaced should show corresponding decline, and they do. Before opening, southern-bound passengers chose overnight launches from Sadarghat or vehicle ferries at Mawa, both imposing heavy time penalties, the launch to Barishal taking roughly nine hours and the ferry adding one to two and a half hours of waiting, far more during festivals (Prothom Alo, 2022). The bridge cut the road journey to three to four hours. Since opening, ferry crossings on the corridor became negligible: the Shimulia-Banglabazar route was discontinued after vessels struck bridge pillars (TBS News, 2023a), and remaining services attract little traffic given documented deficits in safety, waiting time, and reliability relative to the bridge (Sourav et al., 2024). The bridge is now the only practical land link to these seventeen districts, so displaced demand necessarily appears on it.

Table 11 quantifies the waterway decline across four indicators from the field surveys and operator records. Launch deck fares, 400 to 600 taka in practice before opening, fell to 150 to 250 by 2025; cabin occupancy fell from 75 to about 35 percent; daily Sadarghat passengers for southern destinations fell from roughly 50,000 to 33,000 within a year (Dhaka Tribune, 2023), a decline the water authority corroborates at 23 to 25 percent (TBS News, 2023b); and daily Barishal-bound departures fell from eight or nine per direction to two by 2025 (Daily Star, 2025), while the state ferry operator reported a 70 percent income collapse (TBS News, 2023a). The roadside absorbed this: the authors' field surveys recorded bus departures on the Dhaka-Barishal corridor growing from approximately 300 to 800 per day at the festival peak and Dhaka-Khulna from approximately 200 to 500, with three operators entering the Barishal market with no prior presence and the state corporation expanding from 18 buses on two routes to 58 to 63 on 23 routes (New Age, 2022). This market entry and fleet expansion is the behavioral counterpart of the toll-recorded growth.

Table 11. Modal-Shift Indicators: Waterway Decline and Road Substitution

Indicator	Pre-bridge	~12 mo	~22 mo	2025
Launch deck fare, Dhaka-Barishal (Tk)	352 (400-600 actual)	—	200	150-250
Launch cabin occupancy (%)	75	50	40	~35
Barishal-bound launches per day	8-9	3-4	4	2
Total Sadarghat departures per day	~140	~60	~60	40-45
Daily launch passengers (all southern)	~50,000	~33,000	—	~20-25k
Dhaka-Barishal buses/day (Eid peak)	~300	—	800	800
Dhaka-Khulna buses/day (Eid peak)	~200	—	500	500

Sources: authors' primary field survey (2022-2024); Dhaka Tribune (2023); TBS News (2023a, 2023b); Daily Star (2025); New Age (2022).

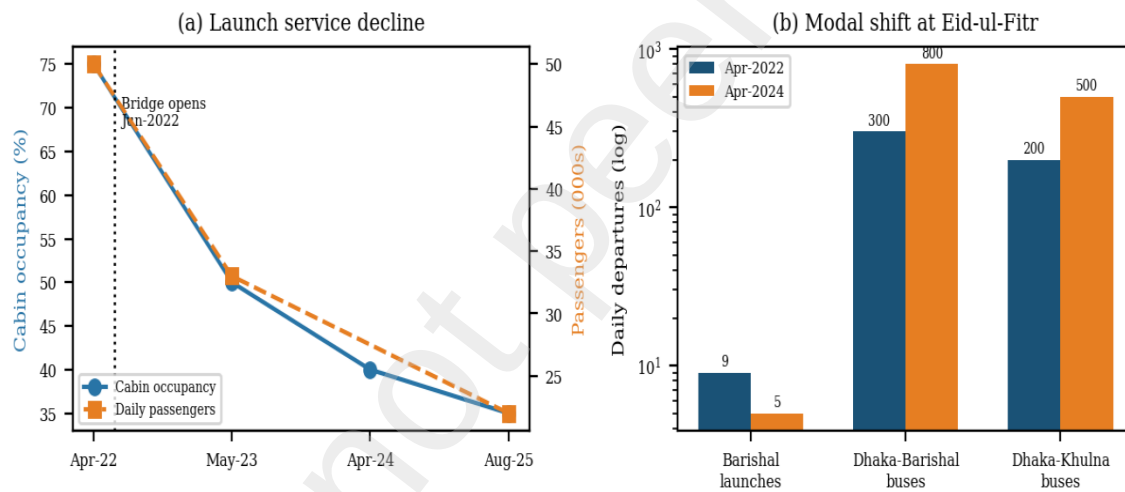


Figure 9. (a) Launch cabin occupancy and daily Sadarghat passenger volumes, both declining after the June 2022 opening. (b) Modal comparison at the festival peak: Barishal launch departures against Dhaka-Barishal and Dhaka-Khulna buses, before and after, on a logarithmic scale.

7 Operational Implications

Because the Bangladesh Bridge Authority publishes booth counts and per-vehicle service times, the capacity analysis rests on the bridge's own operating data. The bridge opened with seven toll booths at the Mawa (Dhaka-side) end and six at Jazira, with service times reported at under one minute per vehicle at Mawa and roughly thirty seconds at Jazira, implying manual throughput of roughly 60–75 vehicles per booth per hour (TBS News, 2022). Since the plazas serve opposite directions, the binding constraint during a festival is single-directional: the Mawa end alone, at about 525 vehicles per hour across seven booths, governs the pre-Eid outbound surge. That the plaza is already at its limit is documented, not projected: peak congestion now occurs on Fridays, holidays, and around Eid, with crossings taking two to four hours (TollGuru, 2026), and the Authority repeatedly opens emergency bays and temporary motorcycle booths yet long queues still form (TBS News, 2025; bdnews24, 2025). On the record day the Mawa end alone handled 28,415 outbound vehicles, and the design-day projection of Section 5.5 shows festival peaks approaching and exceeding this. The measures below follow from this condition and the demand asymmetry of Section 5.4.

7.1 Expand Throughput: Electronic Conversion First, New Bays if Needed

The highest-leverage step is to replicate electronic (radiofrequency) toll collection across as many existing booths as physically possible at each plaza, keeping only one or two cash lanes per end for vehicles without transponders. A manual booth clearing 75 vehicles per hour becomes, as a free-flow electronic lane, capable of 300 to 500 per hour on the very same bay, a four- to sixfold gain that requires no construction. Electronic collection already exists at the bridge but carries only a minority of traffic, so the task is to convert most bays and drive transponder uptake, capturing the gain where the concrete already stands; this is the only lever that scales once the built bays are fully staffed. Where even full conversion cannot close the gap, new bays should be added at the Mawa end, since the plaza was built with more bays than are normally manned and the Authority already exhausts that reserve during festivals. Because festival demand keeps rising, with the record climbing from 51,316 vehicles in 2022 to 52,487 in 2025 and the design-day projection reaching similar levels by 2028, the Mawa end will require on the order of 4,000 vehicles per hour to stay within accepted service levels, which seven converted bays may not reach; planning two to four additional Mawa bays now would avoid the emergency improvisation that recurs every festival.

7.2 Schedule Capacity by Direction and Festival Timing

Deployment should be scheduled by direction and by the pre- and post-festival asymmetry this study quantified, rather than triggered reactively once queues form. Because outbound demand in the three days before Eid (35,991 vehicles per day) exceeds return demand in the three days after (27,729), the rule is specific: staff every bay and concentrate electronic and motorcycle lanes at the Mawa end during the three days before each Eid, then shift that concentration to Jajira for the five days after, as the flow reverses. Since Eid dates are fixed by the Islamic calendar more than a year ahead and the directional pattern is now documented, this reallocation can be pre-scheduled at zero marginal cost, turning reactive practice into a calendar-driven plan that positions capacity ahead of the surge and on the correct side of the river.

7.3 Make Motorcycle Provision Permanent

Motorcycles, at 17.5 percent of traffic in the ground-truth window and up to a third of Mawa-end flow on peak festival days, are currently served by booths that swing from one in normal periods to three during festivals, with no permanent dedicated lane on the approach. A permanent dedicated motorcycle lane on the Mawa approach, together with permanently assigned motorcycle booths sized to the observed share, would lift the largest single vehicle class out of the mixed queue and ease interference across all remaining lanes.

8 Conclusions

This study presented the first peer-reviewed post-inauguration traffic verification for a South Asian megaproject bridge, using 1,442 days of official toll data to compare statistical and deep learning forecasts and to benchmark observed demand against the projections that justified the investment.

The central methodological finding is that the SARIMAX and LSTM model classes are statistically indistinguishable on this series: the LSTM attained lower holdout error, SARIMAX won on cross-validated mean error and on bias, and the Diebold-Mariano test could not reject equal accuracy ($p = 0.109$). Model selection at this data scale should therefore rest on interpretability, determinism, and unbiasedness rather than point-error differences within sampling variation, and SARIMAX's transparency and near-zero bias recommend it for revenue-relevant forecasting, while the LSTM's run-to-run variability proved smaller than the literature warns. The central substantive finding is that both models degrade by factors of 2.7 to 3.9 within festival windows, traced by the diagnostics to a demand process that is heavy-tailed and, more importantly, asymmetric, with pre-festival traffic exceeding post-festival traffic by 30 percent. A symmetric binary indicator is the wrong shape for this process, and a signed directional regressor, available in advance from the Islamic calendar, is the indicated remedy and the most promising extension of this work.

A design-day projection, built as a seasonal scenario rather than a daily-model extrapolation because such models are short-horizon instruments, indicates festival peak days approaching and potentially exceeding the current record of

52,487 vehicles within two years. This directly motivates the operational recommendations: electronic conversion of most toll bays with new bays where needed, directionally scheduled festival capacity keyed to the pre-festival surge, and permanent motorcycle provision, each justified on observed volumes alone. Independent modal-shift evidence from primary field surveys and operator records, showing launch occupancy falling from 75 to about 35 percent and bus services growing 150 to 167 percent on the same corridor, confirms that the forecast demand is genuine diverted demand rather than modeling artifact.

Three limitations bound these conclusions and each defines an opportunity. The apparent year-over-year escalation in festival magnitude could not be confirmed on nine events ($p = 0.173$) and warrants continued monitoring as the record lengthens. The Year 3 and Year 4 level indicators were structurally inestimable within the training block, forcing extrapolation rather than estimation of the mature-phase level, which a longer training window would resolve. And the residual autocorrelation test failed marginally at lag 21 while passing at adjacent lags, leaving open a possible subtle three-week dependency whose source we could not identify. None affects the core comparative result, and each is a natural subject for the continued verification this class of study makes possible.

The framework transfers directly to any tolled megaproject crossing with daily records and a migrating-holiday demand structure. The stationarity protocol, the significance-tested model comparison, the directional festival diagnosis, and the feasibility-forecast benchmarking require only data that most toll authorities already collect, and the modal-shift validation demonstrates how independent operator data can confirm that forecast demand is real. As post-inauguration records accumulate across South and Southeast Asia, systematic verification of the kind demonstrated here can replace the forecast-and-forget practice that currently characterizes megaproject planning.

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Declaration of Generative AI in Scientific Writing

During the preparation of this work, the authors used ChatGPT to polish the writing and improve the clarity of section headings and used Claude (Anthropic) as a coding and analysis assistant for Python scripting, figure generation, and drafting support. All model specifications, analytical decisions, interpretations, and conclusions are the authors' own; the authors reviewed and edited all outputs, verified numerical results and take full responsibility for the content of the published article.

Author Contribution

The authors confirm contributions as follows. M. S. N. Chowdhury: study conception, data collection, modeling, analysis, and manuscript writing. M. M. Rahman: supervision, methodology guidance, and critical review. S. S. Pruma: data collection, field survey work, and review. A S M Shanawaz Uddin: review and editing. All authors reviewed the results and approved the final manuscript.

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